

Wavefunction Energies in an Infinite Square Well

Kieron Eisma

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Introduction

This paper aims to provide a quick derivation of the energy levels of a particle of mass m contained in an infinite square well. Numerous online articles do the same thing; for instance, see [Wik26] and [DA122]. I have therefore opted to provide a thorough treatment of the problem in this article for new students of Quantum Mechanics.

Suggestions and errors are to be forwarded to my email:

kiwi.eisma@gmail.com

1 Schrödinger Equation

Note that, in an infinite square well ($V(0) = V(a) = \infty$), the potential (and hence the system's Hamiltonian) is time-independent by definition. We therefore apply the time-independent Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x).$$

1.1 Outside the Well

Notice how the equation above forces $\psi(x) = 0$ outside the well; otherwise, the particle would possess infinite energy. This implies that the particle is completely contained inside the region between $x = 0$ and $x = a$ (this will be important later when we normalise ψ in Section 2).

1.2 Inside the Well

We have that $V(x) = 0$, so the Schrödinger Equation simplifies to yield

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E\psi(x),$$

or equivalently,

$$\frac{d^2}{dx^2} \psi(x) + \frac{2mE}{\hbar^2} \psi(x) = 0.$$

This is the familiar harmonic differential equation we are used to.¹

We may, therefore, immediately ascertain the solution as

$$\psi(x) = A \cos(kx) + B \sin(kx),$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad (1)$$

Hence, since the wave function vanishes at the walls of this well, we have the conditions $\psi(0) = \psi(a) = 0$. The first condition forces $A = 0$, and the second boundary condition shows that $B \sin ka = 0$. This is only true if either $B = 0$ or $\sin ka = 0$. Now, we recall ψ must satisfy

$$\int_{\mathbb{R}} |\psi(x)|^2 dx = 1,^2$$

So we cannot have $B = 0$. It follows that $\sin ka = 0$, which implies that $ka = n\pi$, for $n \in \mathbb{N}$. In particular,

$$k = \frac{n\pi}{a}, \quad n = 1, 2, \dots$$

Substituting our relation for k in Equation 1 into the one above yields

$$\sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a}.$$

Rearranging for the energy:

¹If you wish to review differential equations of the form $\ddot{x} + \omega^2 x = 0$, see, for instance, Section 4.1 in [Mor08]. For more rigour, consult Chapter 1 of [SS03].

²Intuitively, this tells you the particle must be *somewhere*.

$$\begin{aligned}
E &= \frac{\pi^2 n^2 \hbar^2}{2ma^2} \\
&= \frac{n^2 h^2}{8ma^2},
\end{aligned}$$

where we have utilised the relation for the reduced Planck constant to eliminate π : $\hbar = \frac{h}{2\pi}$.

2 Overall Wave Function

We must subject ψ to the constraint

$$\int_{\mathbb{R}} |\psi(x)|^2 dx = 1,$$

as discussed in Footnote 2. Observe

$$\begin{aligned} \int_{\mathbb{R}} |\psi(x)|^2 dx &= \int_{-\infty}^0 |\psi(x)|^2 dx + \int_0^a |\psi(x)|^2 dx + \int_a^{\infty} |\psi(x)|^2 dx \\ &= \int_0^a |\psi(x)|^2 dx = \int_0^a B^2 \sin^2\left(\frac{\pi nx}{a}\right) dx \\ &= B^2 \int_0^a \sin^2\left(\frac{\pi nx}{a}\right) dx \\ &= \frac{B^2}{2} \int_0^a \left(1 - \cos\left(\frac{2\pi nx}{a}\right)\right) dx \\ &= \frac{B^2}{2} \left[x - \frac{a}{2\pi n} \sin\left(\frac{2\pi nx}{a}\right) \right]_0^a \\ &= \frac{B^2 a}{2} := 1, \end{aligned}$$

where the tail integrals $\int_{-\infty}^0 |\psi(x)|^2 dx$ and $\int_a^{\infty} |\psi(x)|^2 dx$ are annihilated because of our conditions on ψ , discussed in Section 1.1.

This implies³

$$B = \sqrt{\frac{2}{a}}, \tag{2}$$

so that we can express ψ neatly as

$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi nx}{a}\right).$$

³I take the positive root for aesthetic reasons.

Here is a nice figure that shows the wave functions ψ_1 through ψ_4 , along with their corresponding energies (derived in Section 2.2) shown to the right. Notice how the higher energy states correspond to a greater frequency of the wave.

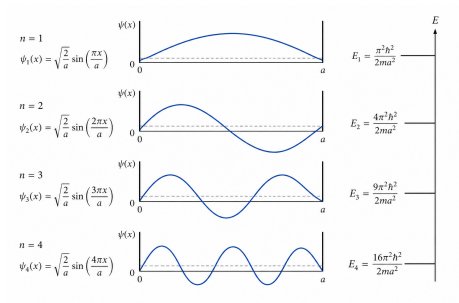


Figure 1: Plot of the first four energy levels. Image generated by ChatGPT 5.5.

References

- [DA122] Peter D'Alessandris. *6.2: Solving the 1D Infinite Square Well*. 2022. URL: [https://phys.libretexts.org/Bookshelves/Modern_Physics/Spiral_Modern_Physics_\(D%27Alessandris\)/6%3A_The_Schrodinger_Equation/6.2%3A_Solving_the_1D_Infinite_Square_Well](https://phys.libretexts.org/Bookshelves/Modern_Physics/Spiral_Modern_Physics_(D%27Alessandris)/6%3A_The_Schrodinger_Equation/6.2%3A_Solving_the_1D_Infinite_Square_Well) (visited on 08/15/2026).
- [Mor08] David Morin. *Introduction to Classical Mechanics: With Problems and Solutions*. Cambridge University Press, 2008.
- [SS03] Elias M. Stein and Rami Shakarchi. *Fourier Analysis: An Introduction*. Vol. 1. Princeton Lectures in Analysis. Princeton University Press, 2003.
- [Wik26] Wikipedia contributors. *Particle in a Box*. 2026. URL: https://en.wikipedia.org/wiki/Particle_in_a_box (visited on 08/15/2026).